Dharm Veer

Curriculum Vitae

Basic Information

Name	Dharm Veer
Occupation	Post-Doctoral Fellow
Institution	Sabancı University, Istanbul, Turkey
e-mail	dharmveer.sharma091@gmail.com
Homepage	https://cmidharm.github.io/

Research Interest

Commutative Algebra, and Combinatorics.

Academic Positions

May 23, 2026 – till date	**Postdoctoral fellow**, *Sabancı University*, Istanbul, Turkey.
May 2, 2024 – April, 30, 2026	**AARMS Postdoctoral fellow**, *Dalhousie University*, Halifax, Canada.
November 25, 2022 – April, 2024	**Postdoctoral fellow**, *Indian Institute of Technology Gandhinagar*, Gandhinagar, India.

Education

2016 – 2023	**Ph.D. in Mathematics**, *Chennai Mathematical Institute*, Chennai, India, Thesis title : Homological Invariants of Hibi Rings and Polyominoes. Thesis advisor : Prof. Manoj Kummini.
2014 – 2016	**Master of Science in Mathematics**, *Chennai Mathematical Institute*, Chennai, India.
2011 – 2014	**Bachelor of Science (Honours) in Mathematics**, *Motilal Nehru College, University of Delhi*, New Delhi, India.
2011	**12th CBSE Board Examination**, Jawahar Navodaya Vidyalaya, Faridabad, India.

Publications/Pre-prints

Publications

[1] Rizwan Jahangir and Dharm Veer. “On Cohen-Macaulay posets of dimension two and permutation graphs”. *Bull. Malays. Math. Sci. Soc.* 48.4 (2025), Paper No. 133, 10.

[2] Dharm Veer. “Green-Lazarsfeld property N_p for Segre product of Hibi rings”. *To appear in Indian j. pure appl. math.* (2024).

[3] P. V. Cheri, Deblina Dey, Akhil K, Nirmal Kotal, and Dharm Veer. “Cohen-Macaulay permutation graphs”. *Math. Scand.* 130.3 (2024), pp. 419–431.

[4] Carmelo Cisto, Francesco Navarra, and Dharm Veer. “Polyocollection ideals and primary decomposition of polyomino ideals”. *J. Algebra* 641 (2024), pp. 498–529.

[5] Dharm Veer. “On the linearity of the syzygies of Hibi rings”. *J. Algebra Appl.* 23.12 (2024), p. 2450208.

[6] Manoj Kummini and Dharm Veer. “The h-polynomial and the rook polynomial of some polyominoes”. *Electron. J. Combin.* 30.2 (2023), P2.6.

[7] Manoj Kummini and Dharm Veer. “The Charney-Davis conjecture for simple thin polyominoes”. *Commun. Algebra* 51.4 (2023), pp. 1654–1662.

Submitted

[1] Mitra Koley, Nirmal Kotal, and Dharm Veer. “Polyominoes and Knutson ideals”. *arXiv preprint arXiv:2411.16364* (2024).

[2] Louis Bu, Sara Faridi, Iresha Madduwe Hewalage, Thiago Holleben, Hasan Mahmood, Dharm Veer, Kyle Wang, and Scott Wesley. “When are Morse resolutions polyhedral?”: *arXiv preprint arXiv:2505.08580* (2025).

[3] Francesco Navarra, Ayesha Asloob Qureshi, and Dharm Veer. “t-Young complexes and squarefree powers of t-path ideals”. *arXiv preprint arXiv:2511.13477* (2025).

[4] Sara Faridi, Dharm Veer, and Volkmar Welker. “Complements and complementary homologies”. *preprint* (2026).

In Preparation

[1] Takayuki Hibi and Dharm Veer. “Polyomino algebras and Hibi rings”. (2025).

[2] Dharm Veer. “Quadratic binomial ideals associated to certain combinatorial objects”. (2025).

Research Talks and seminar presentations

- Complements and complementary homologies, *Fourth Meru Combinatorics Conference 2026*, IIT Bhilai, Bhilai, India. June 2026.
- Complements and complementary homologies, *Algebra Seminar*, McMaster University, Hamilton, Canada. March 2026.
- Free resolution of squarefree powers of simplicial trees, *SLMath Special Session on Combinatorial Aspects in Commutative Algebra*, Joint Mathematics Meetings (JMM) 2026, Washington, D.C, USA. January 2026.
- Binomial ideals associated to polycubes, *Commutative Algebra Session*, CMS Winter Meeting 2025, Toronto, Canada. December 2025.
- Delivered two lectures on *Characteristic-dependent Betti numbers of monomial ideals and their powers*, Commutative Algebra Seminar, Dalhousie University, Halifax. October, 2024.
- Delivered two lectures on *Characteristic-dependent Betti numbers of monomial ideals and their powers*, Commutative Algebra Seminar, Dalhousie University, Halifax. October, 2024.
- Delivered two lectures on *Grothendieck Quot Scheme and Dimension estimates*, Graduate Seminar series on Moduli space of sheaves, IIT Gandhinagar, Gandhinagar. August-September, 2023.
- On EL-shellable posets, Workshop on *Cohen Macaulay simplicial complexes in graph theory*, Chennai Mathematical Institute, Chennai, India. July, 2023.
- On Cohen-Macaulay posets of dimension 2 and permutation graphs, GANIT Seminar, IIT Gandhinagar, India. March 2023.
- On Cohen-Macaulay posets of dimension 2 and permutation graphs, Conference on Commutative Algebra and Algebraic Geometry, IIT Hyderabad, India. February 2023.

- Polyominoes and polyomino ideals, Graduate Seminar, IIT Gandhinagar, India. January 2023.
- On the h-polynomial and the rook polynomial of some polyominoes, EMS Summer School on Combinatorial Commutative Algebra, Gebze Technical University, Istanbul, Turkey. August 2022.
- On Green-Lazarsfeld property N_p for Hibi rings, Virtual Commutative Algebra Seminars, IIT Bombay, Mumbai. March 2022. Video. Notes.
- On Green-Lazarsfeld property N_p for Hibi rings, National symposium on mathematics and applications, IIT Madras, Chennai. December 2021.
- On Green-Lazarsfeld property N_p for Hibi rings, 4^{th} BRICS Mathematics Conference, IISER Tiruvananthapuram, Kerala. December 2021. Video.
- On Green-Lazarsfeld property N_p for Hibi rings, Research Seminar, CMI, Chennai. September, 2021.
- On h-Polynomials of Hibi rings, online talk in webinar Series: CATGT. August 2020. Video.
- Delivered three lectures on *Positive Characteristic Commutative Algebra*, Commutative Algebra Seminar, CMI, Chennai. January, 2020.
- Delivered two lectures on *Tate resolutions*, Commutative Algebra Seminar, CMI, Chennai. November, 2019.
- On h-Polynomials of Hibi rings, Research Seminar, CMI, Chennai. September, 2019.
- Delivered three lectures on *Homological Invariants of Modules*, Commutative Algebra Seminar, CMI, Chennai. April, 2019. Based on Prof. Srikanth Iyengar's lectures.
- Absolute Integral Closure In Positive Characteristic, Mates PhD Research Seminar, CMI, Chennai. March, 2017.

Teaching

- MATH 2505: Introductory Real Analysis. Dalhousie University, Canada. January-April, 2025.

Teaching Assistantships

- MA104: Ordinary Differential Equations. IIT Gandhinagar, India. January-February, 2024.
  Course Instructor : Prof. Jagmohan Tyagi.
- MA103: Calculus of Single Variable and Linear Algebra. IIT Gandhinagar, India. August-November, 2023.
  Course Instructor : Prof. Projesh Nath Choudhury.
- Computational Commutative Algebra, NPTEL Online course. July-November, 2022.
  Course Instructor : Prof. Manoj Kummini.
- Computational Commutative Algebra, NPTEL Online course. July-November, 2021.
  Course Instructor : Prof. Manoj Kummini.
- Computational Commutative Algebra, NPTEL Online course. September-December, 2020.
  Course Instructor : Prof. Manoj Kummini.
- Algebra I, Chennai Mathematical Institute, Chennai, India. August-November, 2019.
  Course Instructor : Prof. Purusottam Rath.
- Madhava Mathematics Competition Nurture Camp, *Invariant theory of finite groups*, CMI, Chennai, India. June 2019.
  Course Instructor : Prof. Manoj Kummini.
- Homological algebra, Chennai Mathematical Institute, Chennai, India. January-April, 2019.
  Course Instructor : Prof. Kumari Saloni.

Research Visits

- IIT Bhilai, Bhilai, India. May, 2026.
  Host : Prof. Anurag Singh.
- University of Genoa, Genoa, Italy. June, 2025.
  Host : Prof. Aldo Conca.
- Sabancı University, Tuzla, Turkey. May, 2025.
  Host : Prof. Ayesha Asloob Qureshi.
- Max Planck Institute for Mathematics, Bonn, Germany. Nov - Dec, 2023.
- Sabancı University, Tuzla, Turkey. August, 2022.
  Host : Prof. Ayesha Asloob Qureshi.

Internships

May 22 – August 6, 2016 **Summer Fellow**, Centre for Excellence in Basic Sciences (CBS), Mumbai, India, Supervisor : Prof. Balwant Singh.

May 3 – June 2, 2015 **Summer Fellow**, Centre for Excellence in Basic Sciences (CBS), Mumbai, India, Supervisor : Prof. Balwant Singh.

Participation in Workshops/Conferences

- Fourth Meru Combinatorics Conference 2026, IIT Bhilai, Bhilai, India. June 2026
- Joint Mathematics Meetings (JMM) 2026, Washington, D.C., USA. January 2026.
- Commutative Algebra Session, CMS Winter Meeting 2025, Toronto, Canada. December 2025.
- Conference on Homological Commutative Algebra and Related Topics, Georgia Southern University, Savannah, USA. August, 2025.
- Modules and Rings: Recent Developments in Commutative Algebra, a conference in honor of Marilina Rossi, University of Genova, Genova, Italy. June, 2025.
- Workshop on *Weak and Strong Lefschetz Properties across Mathematics*, Sophus Lie Conference Center, Nordfjordeid, Norway. June, 2025.
- Conference on *Combinatorial Algebra Meets Algebraic Combinatorics (CAAC)*, The Fields Institute, Toronto, Canada. January, 2025.
- Apprenticeship Program in Commutative Algebra, The Fields Institute, Toronto, Canada. January, 2025.
- Workshop on *Cohen Macaulay simplicial complexes in graph theory*, Chennai Mathematical Institute, Chennai, India. July, 2023.
- Conference on Commutative Algebra and its interaction with Algebraic Geometry and Combinatorics, VIASM Hanoi, Vietnam. June 2023.
- AIS School on *Topics in Birational Geometry*, Chennai Mathematical Institute, Chennai, India. May-June, 2023.
- Conference on Commutative Algebra and Algebraic Geometry, IIT Hyderabad, India. February 2023.
- EMS Summer School on Combinatorial Commutative Algebra, Gebze Technical University, Istanbul, Turkey. August, 2022.
- A Conference on Topics in Algebraic Geometry and Commutative Algebra, SRM University-AP, Andhra Pradesh, India. July, 2022.

- NCM Workshp on *maximal Cohen-Macaulay Modules*, Chennai Mathematical Institute, Chennai, India. July, 2022.
- 4^{th} BRICS Mathematics Conference, IISER Tiruvananthapuram, Kerala, India. December, 2021.
- AIS School on *advanced commutative algebra*, IIT Kharagpur, India. December, 2019.
- Workshop on Hochschild Homology, Chennai Mathematical Institute, Chennai, India. July, 2019.
- NCM Workshop *commutative algebra and algebraic geometry in positive characteristics*, IIT Bombay, Mumbai, India. December, 2018.
- Projective Modules, IIT Bombay, Mumbai, India. May-June, 2018.
- AIS School on *Gröbner Bases and their Applications*, IIIT Delhi, New Delhi, India. December, 2017.
- ATM Workshop on *positive Characteristic Methods in Commutative Algebra*, IIT Bombay, Mumbai, India. June, 2017.
- Workshop on Seshadri constants, Chennai Mathematical Institute, Chennai, India. February, 2017.
- AIS School on commutative Algebra, Chennai Mathematical Institute, Chennai, India. December, 2015.

Awards and Achievements

May 3, 2024 – April 30, 2026 **AARMS Postdoctoral fellowship.**
Awarded by AARMS, Canada.

August, 2018 – November, 2022 **Senior Research Fellowship.**
Awarded by Chennai Mathematical Institute, Chennai, India.

August, 2016 – July, 2018 **Junior Research Fellowship.**
Awarded by Chennai Mathematical Institute, Chennai, India.

August, 2014 – April, 2016 **Postgraduate Scholarship.**
Awarded by Chennai Mathematical Institute, Chennai, India.

2016 **Selected for PhD in IISER Pune (didn't avail)..**

2016 **Qualified Graduate Apptitude Test in Engineering(GATE).**
Conducted by Indian Institute of Technology (IIT).

December 2015 **73rd Rank in CSIR-JRF Dec-2015 Mathematical Science test.**
Conducted by CSIR-UGC for Junior Research Fellowship and eligibility for lectureship.

December 2014 **44th Rank in CSIR-NET Dec-2014 Mathematical Science test.**
Conducted by CSIR-UGC for eligibility for lectureship.

2014 **62nd Rank in IIT – JAM 2014 Mathematics subject test.**
Conducted by Indian Institute of Technology (IIT) for Master's programme in Mathematics.

2004 – 2011 **Free Education and Accommodation From Class 6th to 12th.**
Awarded by HRD Ministry of India on the Basis of Aptitude Test.

Other Academic Activities

- **Co-organizer (with Graham Denham, Sara Faridi , Selvi Kara, and Hugh Thomas)**, *of the Combinatorial Algebra meets Algebraic Combinatorics (CAAC) 2026 Conference*, at Dalhousie University, Halifax, Canada. $23-25$ January 2026.

- **Organizer of Weekly Commutative Algebra Seminar**, *at Dalhousie University, Halifax, Canada. Since October* 2024.